

LAB4

Exercise 36

36. Spray drift is a constant concern for pesticide applicators and agricultural producers. The inverse relationship between droplet size and drift potential is well known. The paper “Effects of 2,4-D Formulation and Quinclorac on Spray Droplet Size and Deposition” (*Weed Technology*, 2005: 1030–1036) investigated the effects of herbicide formulation on spray atomization. A figure in the paper suggested the normal distribution with mean $1050 \mu\text{m}$ and standard deviation $150 \mu\text{m}$ was a reasonable model for droplet size for water (the “control treatment”) sprayed through a 760 ml/min nozzle.
- What is the probability that the size of a single droplet is less than $1500 \mu\text{m}$? At least $1000 \mu\text{m}$?
 - What is the probability that the size of a single droplet is between 1000 and $1500 \mu\text{m}$?
 - How would you characterize the smallest 2% of all droplets?
 - If the sizes of five independently selected droplets are measured, what is the probability that at least one exceeds $1500 \mu\text{m}$?

Solution

We have $X \sim N(1050, 150^2)$

$$\text{a) } P(X < 1500) = P\left(Z < \frac{1500 - 1050}{150}\right) = P(Z < 3) = \mathbf{0.9987}$$

$$P(X \geq 1000) = 1 - P(X < 1000) = 1 - P\left(Z < \frac{1000 - 1050}{150}\right) \approx 1 - P(Z < -0.33) \approx 1 - 0.3707 = \mathbf{0.6293}$$

$$\text{b) } P(1000 \leq X \leq 1500) = P(X \leq 1500) - P(X < 1000) = 0.9987 - 0.3707 = \mathbf{0.628}$$

c) Let the size be x

$$\text{We have } P\left(Z < \frac{x - 1050}{150}\right) = 0.02 \Rightarrow \frac{x - 1050}{150} \approx -2.055 \Rightarrow x \approx \mathbf{741.75}$$

This states that the smallest 2% of droplets falls under $741.75 \mu\text{m}$.

$$\text{d) From a, we have } P(X > 1500) = 1 - 0.9987 = 0.0013$$

Let $Y \sim B(5, 0.0013)$

$$P(Y \geq 1) = 1 - P(Y < 1) = 1 - P(Y = 0) = 1 - C_5^0 \times 0.0013^0 \times (1 - 0.0013)^{5-0} \approx \mathbf{0.0065}$$

Exercise 37

37. Suppose that blood chloride concentration (mmol/L) has a normal distribution with mean 104 and standard deviation 5 (information in the article “Mathematical Model of Chloride Concentration in Human Blood,” *J. of Med. Engr. and Tech.*, 2006: 25–30, including a normal probability plot as described in Section 4.6, supports this assumption).
- What is the probability that chloride concentration equals 105? Is less than 105? Is at most 105?
 - What is the probability that chloride concentration differs from the mean by more than 1 standard deviation? Does this probability depend on the values of μ and σ ?
 - How would you characterize the most extreme .1% of chloride concentration values?

Solution

We have $X \sim N(104, 5^2)$

a) $P(X=105) = 0$

$$P(X < 105) = P(X \leq 105) = P\left(Z \leq \frac{105-104}{5}\right) = P(Z \leq 0.2) = \mathbf{0.5793}$$

b) $P(|X - \mu| > \sigma) \Leftrightarrow P(X - \mu > \sigma) + P(X - \mu < -\sigma) (*)$

$$\text{Let } Z = \frac{X - \mu}{\sigma}$$

$$(*) \Leftrightarrow P(Z > 1) + P(Z < -1) = 1 - P(Z \leq 1) + P(Z < -1) = 1 - 0.8413 + 0.1587 = \mathbf{0.3174}$$

Hence, this probability **doesn't depend on** the values of μ and σ

c) This equivalent to $P(X_1 < c_1) = 0.0005$ and $P(X_1 > c_2) = 0.0005$

$$P(X_1 < c_1) = 0.0005 \Leftrightarrow P\left(Z < \frac{c_1 - 104}{5}\right) = 0.0005 \Rightarrow \frac{c_1 - 104}{5} \approx -3.295 \Rightarrow c_1 \approx 87.525$$

$$P(X_1 > c_2) = 0.0005 \Leftrightarrow 1 - P\left(Z \leq \frac{c_2 - 104}{5}\right) = 0.0005 \Rightarrow \frac{c_2 - 104}{5} \approx 3.295 \Rightarrow c_2 \approx 120.475$$

Hence, the most extreme 0.1% of chloride concentration values is **< 87.525** or **> 120.475**

Exercise 40

40. The article “Monte Carlo Simulation—Tool for Better Understanding of LRFD” (*J. Structural Engr.*, 1993: 1586–1599) suggests that yield strength (ksi) for A36 grade steel is normally distributed with $\mu = 43$ and $\sigma = 4.5$.
- What is the probability that yield strength is at most 40? Greater than 60?
 - What yield strength value separates the strongest 75% from the others?

Solution

We have $X \sim N(43, 4.5^2)$

$$\text{a) } P(X \leq 40) = P\left(Z \leq \frac{40 - 43}{4.5}\right) \approx P(Z \leq -0.67) = \mathbf{0.2514}$$

$$P(X > 60) = 1 - P(Z \leq 60) = 1 - P\left(Z \leq \frac{60 - 43}{4.5}\right) \approx 1 - P(Z \leq 3.78) \approx \mathbf{0.0001}$$

b) Let the value be x

$$\text{We have } P\left(Z > \frac{x - 43}{4.5}\right) = 0.75 \Leftrightarrow 1 - P\left(Z \leq \frac{x - 43}{4.5}\right) = 0.75 \Rightarrow \frac{x - 43}{4.5} \approx -0.675 \Rightarrow x \approx \mathbf{39.9625}$$

Hence, yield strength of approximately 39.925 separates the strongest 75% from the others

Exercise 48

48. Let $X_1, X_2, \dots, X_{100}$ denote the actual net weights of 100 randomly selected 50-lb bags of fertilizer.

a. If the expected weight of each bag is 50 and the variance is 1, calculate $P(49.9 \leq \bar{X} \leq 50.1)$ (approximately) using the CLT.

b. If the expected weight is 49.8 lb rather than 50 lb so that on average bags are underfilled, calculate $P(49.9 \leq \bar{X} \leq 50.1)$.

Solution

$$\text{a) We have } \bar{X} \sim N\left(50, \frac{1}{100}\right)$$

$$P(49.9 \leq \bar{X} \leq 50.1) = P\left(\frac{49.9 - 50}{\sqrt{1/100}} \leq Z \leq \frac{50.1 - 50}{\sqrt{1/100}}\right) = P(-1 \leq Z \leq 1) = P(Z \leq 1) - P(Z < -1) = \mathbf{0.6826}$$

$$\text{b) We have } \bar{X} \sim N\left(49.8, \frac{1}{100}\right)$$

$$P(49.9 \leq \bar{X} \leq 50.1) = P\left(\frac{49.9 - 49.8}{\sqrt{1/100}} \leq Z \leq \frac{50.1 - 49.8}{\sqrt{1/100}}\right) = P(1 \leq Z \leq 3) = P(Z \leq 3) - P(Z < 1) = \mathbf{0.1574}$$

Exercise 52

52. The lifetime of a certain type of battery is normally distributed with mean value 10 hours and standard deviation 1 hour. There are four batteries in a package. What lifetime value is such that the total lifetime of all batteries in a package exceeds that value for only 5% of all packages?

Solution

Mean and standard deviation of a package is:

$$\mu_p = 4 \times \mu = 4 \times 10 = 40$$

$$\sigma_p = \sqrt{4} \times \sigma = 2 \times 1 = 2$$

We have $X_p \sim N(40, 2^2)$

Let the value be x

$$\text{We have } P(Z_p > \frac{x-40}{2}) = 0.05 \Leftrightarrow 1 - P(Z_p \leq \frac{x-40}{2}) = 0.05 \Rightarrow \frac{x-40}{2} \approx 1.645 \Rightarrow x \approx \mathbf{43.29}$$

Hence, lifetime value of approximately 43.29 hours separates the 5% of longest lifetime of all packages

Exercise 59

- 59.** Let X_1 , X_2 , and X_3 represent the times necessary to perform three successive repair tasks at a certain service facility. Suppose they are independent, normal rv's with expected values μ_1 , μ_2 , and μ_3 and variances σ_1^2 , σ_2^2 , and σ_3^2 , respectively.
- If $\mu_1 = \mu_2 = \mu_3 = 60$ and $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = 15$, calculate $P(X_1 + X_2 + X_3 \leq 200)$. What is $P(150 \leq X_1 + X_2 + X_3 \leq 200)$?
 - Using the μ_i s and σ_i s given in part (a), calculate $P(55 \leq \bar{X})$ and $P(58 \leq \bar{X} \leq 62)$.
 - Using the μ_i s and σ_i s given in part (a), calculate $P(-10 \leq X_1 - .5X_2 - .5X_3 \leq 5)$.
 - If $\mu_1 = 40$, $\mu_2 = 50$, $\mu_3 = 60$, $\sigma_1^2 = 10$, $\sigma_2^2 = 12$, and $\sigma_3^2 = 14$, calculate $P(X_1 + X_2 + X_3 \leq 160)$ and $P(X_1 + X_2 \geq 2 X_3)$.

Solution

a) Let $T = X_1 + X_2 + X_3$

We have $\mu_T = \mu_1 + \mu_2 + \mu_3 = 60 + 60 + 60 = 180$

$$\begin{aligned}\sigma_T^2 &= V(X_1 + X_2 + X_3) = V(X_1) + V(X_2) + V(X_3) \text{ (independent variables)} \\ &= \sigma_1^2 + \sigma_2^2 + \sigma_3^2 = 15 + 15 + 15 = 45\end{aligned}$$

$\Rightarrow T \sim N(180, 45)$

$$P(T \leq 200) = P(Z \leq \frac{200-180}{\sqrt{45}}) \approx P(Z \leq 2.98) = \mathbf{0.9986}$$

$$P(150 \leq T \leq 200) = P(T \leq 200) - P(T < 150) \approx P(Z \leq 2.98) - P(Z \leq \frac{150-180}{\sqrt{45}})$$

$$\approx P(Z \leq 2.98) - P(Z \leq -4.47) \approx \mathbf{0.9986}$$

b) We have:

$$\mu_{\bar{X}} = \frac{\mu_1 + \mu_2 + \mu_3}{3} = \frac{180}{3} = 60$$

$$\begin{aligned}\sigma_{\bar{X}}^2 &= V\left(\frac{X_1 + X_2 + X_3}{3}\right) = \frac{1}{9}[V(X_1) + V(X_2) + V(X_3)] \text{ (independent variables)} \\ &= \frac{\sigma_1^2 + \sigma_2^2 + \sigma_3^2}{9} = \frac{45}{9} = 5\end{aligned}$$

$$\Rightarrow \bar{X} \sim N(60, 5)$$

$$P(55 \leq \bar{X}) = 1 - P(\bar{X} < 55) = 1 - P(Z < \frac{55-60}{\sqrt{5}}) \approx 1 - P(Z < -2.24) = \mathbf{0.9875}$$

$$P(58 \leq \bar{X} \leq 62) = P(\bar{X} \leq 62) - P(\bar{X} < 58) = P(Z \leq \frac{62-60}{\sqrt{5}}) - P(Z < \frac{58-60}{\sqrt{5}})$$

$$\approx P(Z \leq 0.89) - P(Z < -0.89) = \mathbf{0.6266}$$

$$\text{c) Let } Y = X_1 - 0.5X_2 - 0.5X_3$$

We have:

$$\mu_Y = \mu_1 - 0.5\mu_2 - 0.5\mu_3 = 60 - 30 - 30 = 0$$

$$\sigma_Y^2 = V(X_1 - 0.5X_2 - 0.5X_3) = V(X_1) + (0.5)^2V(X_2) + (0.5)^2V(X_3) \text{ (independent variables)}$$

$$= \sigma_1^2 + 0.25\sigma_2^2 + 0.25\sigma_3^2 = 15 + 0.25 \times 15 + 0.25 \times 15 = 22.5$$

$$\Rightarrow Y \sim N(0, 22.5)$$

$$P(-10 \leq Y \leq 5) = P(Y \leq 5) - P(Y < -10) = P(Z \leq \frac{5-0}{\sqrt{22.5}}) - P(Z < \frac{-10-0}{\sqrt{22.5}})$$

$$\approx P(Z \leq 1.05) - P(Z < -2.11) = \mathbf{0.8357}$$

d)

$$\text{i) Let } M = X_1 + X_2 + X_3$$

$$\text{We have } \mu_M = \mu_1 + \mu_2 + \mu_3 = 40 + 50 + 60 = 150$$

$$\sigma_M^2 = V(X_1 + X_2 + X_3) = V(X_1) + V(X_2) + V(X_3) \text{ (independent variables)}$$

$$= \sigma_1^2 + \sigma_2^2 + \sigma_3^2 = 10 + 12 + 14 = 36$$

$$\Rightarrow M \sim N(150, 36)$$

$$P(M \leq 160) = P(Z \leq \frac{160-150}{\sqrt{36}}) \approx P(Z \leq 1.67) = \mathbf{0.9525}$$

$$\text{ii) Let } N = X_1 + X_2 - 2X_3$$

$$\text{We have } \mu_N = \mu_1 + \mu_2 - 2\mu_3 = 40 + 50 - 2 \times 60 = -30$$

$$\sigma_N^2 = V(X_1 + X_2 - 2X_3) = V(X_1) + V(X_2) + 2^2 \times V(X_3) \text{ (independent variables)}$$

$$= \sigma_1^2 + \sigma_2^2 + 4 \times \sigma_3^2 = 10 + 12 + 4 \times 14 = 78$$

$$\Rightarrow N \sim N(-30, 78)$$

$$P(X_1 + X_2 \geq 2X_3) = P(X_1 + X_2 - 2X_3 \geq 0) = 1 - P(Z < \frac{0-(-30)}{\sqrt{78}}) \approx 1 - P(Z \leq 3.4) = \mathbf{0.0003}$$

Exercise 65

65. Suppose that when the pH of a certain chemical compound is 5.00, the pH measured by a randomly selected beginning chemistry student is a random variable with mean 5.00 and standard deviation .2. A large batch of the compound is subdivided and a sample given to each student in a morning lab and each student in an afternoon lab. Let $\bar{X}$ = the average pH as determined by the morning students and $\bar{Y}$ = the average pH as determined by the afternoon students.
- If pH is a normal variable and there are 25 students in each lab, compute $P(-.1 \leq \bar{X} - \bar{Y} \leq .1)$. [Hint: $\bar{X} - \bar{Y}$ is a linear combination of normal variables, so is normally distributed. Compute $\mu_{\bar{X}-\bar{Y}}$ and $\sigma_{\bar{X}-\bar{Y}}$.]
 - If there are 36 students in each lab, but pH determinations are not assumed normal, calculate (approximately) $P(-.1 \leq \bar{X} - \bar{Y} \leq .1)$.

Solution

$$\mu = 5$$

$$\sigma = 0.2$$

a) $n = 25$. We have:

$$\mu_{\bar{X}} = \mu_{\bar{Y}} = \mu = 5$$

$$\sigma_{\bar{X}}^2 = \sigma_{\bar{Y}}^2 = \sigma^2 / n = 0.2^2 / 25 = 0.0016$$

$$\text{Hence, } \mu_{\bar{X}-\bar{Y}} = \mu_{\bar{X}} - \mu_{\bar{Y}} = 5 - 5 = 0$$

$$\sigma_{\bar{X}-\bar{Y}} = \sqrt{\sigma_{\bar{X}}^2 + \sigma_{\bar{Y}}^2} = \sqrt{0.0016 + 0.0016} \approx 0.0566$$

$$\text{Let } T = \bar{X} - \bar{Y}$$

$$\Rightarrow T \sim N(0, 0.0566^2)$$

$$P(-0.1 \leq T \leq 0.1) = P(T \leq 0.1) - P(T < -0.1) = P\left(Z \leq \frac{0.1 - 0}{0.0566}\right) - P\left(Z < \frac{-0.1 - 0}{0.0566}\right)$$

$$\approx P(Z \leq 1.77) - P(Z < -1.77) = \mathbf{0.9232}$$

b) Repeat the same steps as in a (because $36 \geq 30$ so by the central limit theorem, we can approximate normal distribution)

$$\text{Let } K = \bar{X} - \bar{Y}$$

$$\Rightarrow K \sim N(0, 0.047^2)$$

$$P(-0.1 \leq K \leq 0.1) = P(K \leq 0.1) - P(K < -0.1) = P\left(Z \leq \frac{0.1 - 0}{0.047}\right) - P\left(Z < \frac{-0.1 - 0}{0.047}\right)$$

$$\approx P(Z \leq 2.13) - P(Z < -2.13) = \mathbf{0.9668}$$